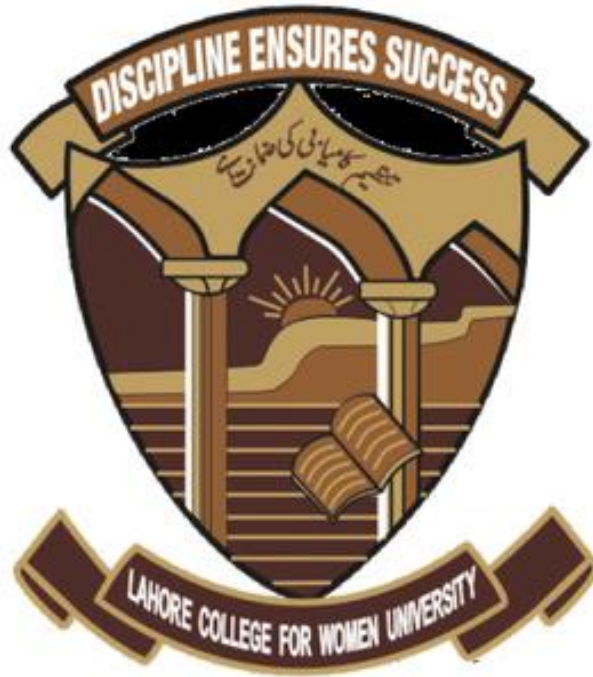


Final Year Project – I (FYP-I)



Department of Software Engineering

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Project Title:

Quizly: An AI-Based Intelligent Quiz and Assessment System with Anti-Cheating Features

Abstract:

The quick rise of online learning has made it more important than ever to have trustworthy and safe digital tools for testing students. But many current online quiz platforms have problems like having to create quizzes by hand, taking a long time to grade essays, not being smart enough to judge answers well, and not having strong ways to stop cheating. These issues make tests unfair, slow, and can affect the honesty of academic work.

This project introduces Quizly, an AI-powered system that helps create and grade quizzes automatically, while also making sure students can't cheat.

The system uses Natural Language Processing to make questions from lecture notes and AI to grade essay-style answers in a fair and consistent way. It also has strong security features like blocking screenshots, detecting when students switch windows, stopping the use of the clipboard, and watching for unusual activity to stop cheating during tests.

In the first part of the project, FYP-I, the main goal is to find the problem, look at what other people have done, analyze the system, design its structure, and plan the project.

The methods, tools, and tech used are decided, and a realistic plan for building the system is created.

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Chapter 1: Introduction

1.1) Background

Online assessment systems have become a key part of modern education because of the rise in e-learning platforms, distance learning, and virtual classrooms. More educational institutions are using digital quizzes and exams to check how well students are performing. While online assessments make learning more flexible and accessible, they also bring up important issues like security, fairness, and the accuracy of evaluations.

Most traditional online quiz platforms mainly use multiple-choice questions and require a lot of manual work for creating and grading tests.

Evaluating answers that require a more personal response takes a lot of time and can be influenced by personal opinions. Also, without good ways to stop cheating, students can sometimes find ways around the system, which harms the honesty of academic work.

New developments in Artificial Intelligence and Natural Language Processing offer a way to fix these problems.

AI-powered systems can handle assessments automatically, cut down on human involvement, and improve fairness. This project uses these technologies to build a safer and smarter assessment tool.

1.2) Problem Statement

Current online testing platforms have various issues. Following are some issues:

- Creating quizzes is done by hand and takes a lot of time.
- There's no automatic way to grade open-ended questions.
- The systems don't have strong ways to stop cheating.
- Feedback is often given late, and evaluations can be unfair.
- These systems also struggle to handle large numbers of students.

All these problems show that there's a need for a smarter, safer, and more automatic way to assess students, making sure that evaluations are fair and honest.

1.3) Objectives

The goals of this project are:

- To look at current online testing systems and figure out what they are missing.
- To create an AI-powered smart quiz tool that can automatically make and mark quizzes.
- To add features that stop students from cheating during tests.
- To build a web app that works well for many users, is easy to use, and allows different roles like teachers and students.
- To help teachers save time and make the grading process faster and better.

1.4) Scope of the Project

Included:

Following features will be including in the project:

- Multiple choice questions, true/false questions, and short-answer questions
- Quiz creation using natural language processing
- Automatic grading for written answers using AI
- Tools to prevent cheating
- Online platform accessible to administrators, teachers, and students

Excluded:

Following features will be excluding and might be including in the future:

- Assessments that take place without an internet connection
- Mobile app
- Advanced proctoring with facial recognition technology

1.5) Significance of the Project

The project has great value from academic, industrial, and social viewpoints, making it a meaningful addition to software engineering and educational technology:

- **Academically:**

From an academic standpoint, the project gives students hands-on experience with cutting-edge technologies like Artificial Intelligence (AI), Natural Language Processing (NLP), and secure web development.

It lets students use their knowledge of machine learning, software design, database systems, and security in a real-life situation. The project also helps advance research in smart assessment systems by tackling issues with current online grading tools, especially when it comes to grading essays and preventing dishonest behavior.

- **Industrially:**

In terms of industry, the system offers a reliable, automated, and affordable way for schools, training centers, and online learning platforms to assess students.

It reduces the workload for teachers, speeds up grading, and ensures fair results without bias. The system uses modern, open-source tools, making it easy to scale up and connect with existing learning platforms.

- **Socially:**

Socially, the project supports fairness and honesty in online learning by stopping cheating and reducing unfair advantages.

It ensures every student has a level playing field and provides quick feedback, which helps improve learning. By supporting honest evaluation, the system builds trust in online education and makes it more accepted by society.

Chapter 2: Literature Review

2.1) Review Of Existing Systems

To understand where online assessment technologies stand today, I looked at several popular platforms and recent research. These tools show what's already available but also point out key shortcomings that make the proposed project necessary.

- **Google Forms:**

Google Forms is a popular choice for creating quizzes and surveys because it's easy to use and accessible.

Teachers can set up simple tests with automatic scoring for multiple-choice questions. But it doesn't have strong security features, which makes it easy for students to cheat. Also, it can't handle subjective answers with AI, so grading those questions still has to be done by hand, which delays feedback.

- **Kahoot:**

Kahoot is another platform that makes quizzes more fun by using game-like elements.

It works well for quick, real-time tests and getting students involved in class. However, it doesn't support grading open-ended questions, secure exam settings, or detailed performance analysis. It is not suitable for official academic testing.

- **Moodle:**

Moodle is a widely used open-source learning management system that supports many assessment types and user roles.

It's more flexible than basic quiz tools but can be hard to set up and manage. Its support for AI-driven grading and anti-cheating tools is limited, often requiring extra plugins or separate software.

- **AI Grading Models:**

Recent research on AI-based assessment models shows promise in using machine learning and natural language processing to grade short-answer and descriptive questions.

These models can help make grading more consistent and reduce bias. But most of these models are still in the testing phase and aren't ready for real-time use, secure environments, or user-friendly setups.

- **Online Proctoring Systems:**

Online proctoring systems use webcams and activity tracking to monitor students during exams.

They help prevent cheating by watching for suspicious behavior. However, they focus more on monitoring than on automating the actual test process. They don't help with creating smart quizzes or grading using AI, and their constant monitoring can be a privacy issue.

In short, current systems handle different parts of assessment like testing, engagement, grading, or security according to their features and systems.

But none of them combine everything into one solution like smart quiz creation, automated grading for written answers, and strong anti-cheating features. This gap is why the proposed project aims to build a single, smart assessment platform that brings all these elements together.

2.2) Comparative Analysis

The comparative analysis of all the current existing online assessment technologies is represented through the table which give the clear difference and analysis for all.

System	Techniques used	Strength	Limitations
Google Forms	They support manual quizzes.	They are easy to use.	It has no security.
Kahoot	They support gamified MCQs.	It engages the users.	There is no subjective grading.
Moodle	It is LMS-based.	It is scalable.	It has a complex setup.
AI Grading Models	It supports Natural Language Processing (NLP) and Machine Learning (ML).	It can do fast grading.	It has limited security.
Online Proctoring Systems	It is based on monitoring the assessment.	It can control cheating.	It has no automation.

2.3) Research Gap

The review of existing online assessment systems and related research reveals a clear gap in the integration of intelligent assessment automation and robust security mechanisms. Most currently available platforms focus either on basic quiz delivery and manual or semi-automated grading, or on monitoring and proctoring to reduce cheating. However, these approaches are typically implemented in isolation rather than as a unified solution.

Existing quiz platforms provide limited automation and primarily support objective-type questions, while subjective answer evaluation remains largely manual and time-consuming. On the other hand, online proctoring systems emphasize surveillance-based monitoring but do not incorporate intelligent quiz generation or AI-based grading techniques. Furthermore, research-oriented AI assessment models demonstrate promising grading accuracy but lack real-time deployment, strong security integration, and scalability for institutional use.

Justification for proposed project:

As a result, there is a significant lack of comprehensive systems that combine AI-based quiz generation, automated subjective grading, and effective anti-cheating mechanisms within a single, user-friendly platform. This research gap highlights the need for an intelligent, secure, and scalable assessment system that ensures fair evaluation, reduces instructor workload, and upholds academic integrity. The proposed project aims to bridge this gap by integrating AI and NLP technologies with advanced anti-cheating features in a unified assessment framework.

Chapter 3: Proposed Methodology / System Design

3.1) System Architecture

The proposed system follows a layered and modular architecture to ensure scalability, security, and maintainability. Each layer is designed to perform a specific set of responsibilities while interacting seamlessly with other components.

- **Frontend:**

The Frontend (User Interface) is responsible for providing an interactive and user-friendly interface for all system users, including administrators, teachers, and students. It handles user input, form validation, quiz navigation, and result visualization. The frontend communicates with the backend through secure API calls.

- **Backend:**

The Backend (Business Logic Layer) manages the core functionality of the system. It handles user authentication, quiz management, question processing, grading logic, and communication with AI and NLP services. This layer ensures data validation, workflow control, and role-based access management.

- **AI and NLP Modules:**

The AI and NLP Modules form the intelligent core of the system. NLP techniques are used to analyze uploaded lecture material and automatically generate questions such as MCQs, True/False, and short-answer questions. AI-based grading models evaluate subjective answers by comparing semantic similarity and contextual relevance, ensuring unbiased and consistent assessment.

- **Database Layer:**

The Database Layer stores all system data, including user information, quizzes, questions, answers, results, and activity logs. MySQL is used for structured and relational data, while MongoDB supports flexible storage for unstructured content such as AI-generated questions and logs.

- **Security and Monitoring Layer:**

The Security and Monitoring Layer enforces system protection and academic integrity. It includes secure authentication, authorization mechanisms, encryption, and real-time activity monitoring. Features such as window-switch detection, screenshot blocking, and clipboard restriction are implemented to minimize cheating during quiz attempts.

3.2) Tools and Technologies

The project utilizes modern, open-source, and widely supported tools and technologies to ensure reliability and scalability.

- **Programming Language:**

JavaScript:

JavaScript is used for both frontend and backend development, enabling seamless full-stack development.

- **Frontend Framework:**

React.js:

React.js is used to build a modular, responsive, and component-based user interface.

- **Backend Framework:**

Node.js with Express.js:

Node.js provides a scalable runtime environment, while Express.js simplifies API development and server-side logic.

- **Database:**

MySQL and MongoDB:

MySQL is used for structured relational data, while MongoDB handles semi-structured and unstructured data.

- **AI and NLP Technologies:**

NLP.js and AI APIs:

These tools enable automated question generation and subjective answer evaluation.

- **Development Tools:**

Visual Studio Code, Postman, GitHub:

These tools support coding, API testing, version control, and collaboration.

3.3) Functional Requirements

The functional requirements define the core features and operations of the proposed system.

- **User Registration and Authentication**

The system shall allow users to register and log in securely with role-based access control for admins, teachers, and students.

- **Quiz Creation and Management**

Teachers shall be able to create quizzes manually or generate them automatically using uploaded lecture materials. Quiz scheduling, time limits, and question randomization shall be supported.

- **Automated Question Generation**

The system shall generate quiz questions using NLP techniques by extracting relevant content from lecture notes.

- **Quiz Attempt and Evaluation**

Students shall be able to attempt quizzes within a secure and monitored environment. Objective questions will be auto-graded, while subjective answers will be evaluated using AI models.

- **Result Generation and Reporting**

The system shall generate instant results and provide performance reports for students and teachers.

3.4) Non-Functional Requirements

The non-functional requirements define the quality attributes of the system.

- **Performance and Responsiveness:**

The system shall respond efficiently under normal and peak loads, ensuring smooth user experience.

- **Security:**

Secure authentication, encrypted data transmission, and anti-cheating mechanisms shall be implemented.

- **Scalability:**

The system shall support an increasing number of users and quizzes without performance degradation.

- **Availability:**

The system shall ensure high availability with minimal downtime.

3.5) Workflow / Use Case Diagram

The system workflow is represented using UML Use Case Diagrams to illustrate interactions between different system users.

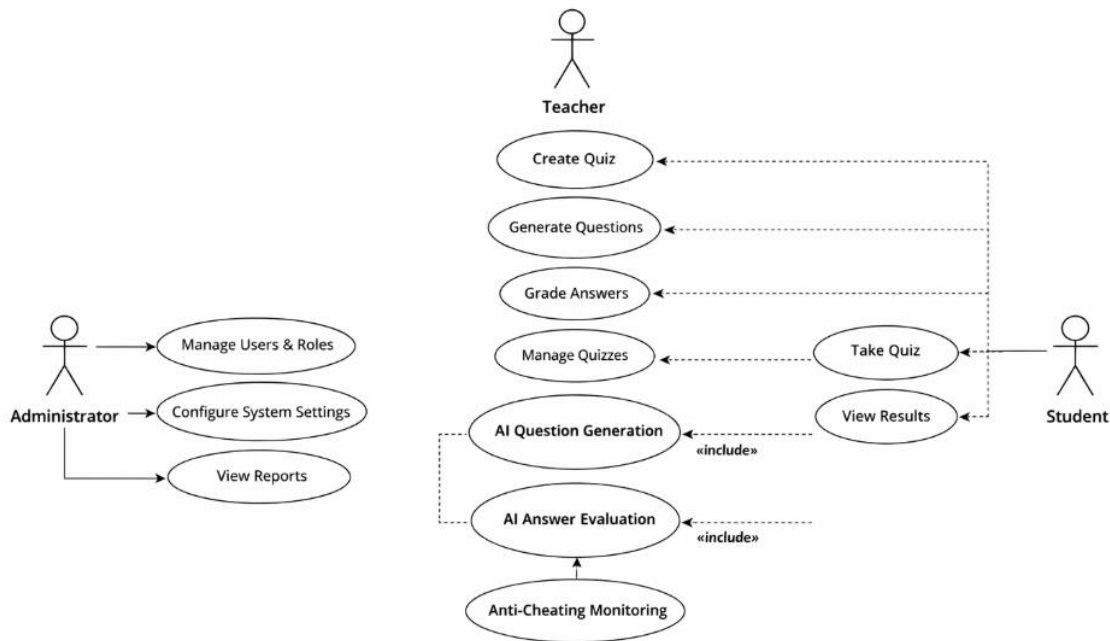
Actors:

- **Administrator:** Manages system users, roles, and configurations.
- **Teacher:** Creates quizzes, uploads lecture material, monitors student performance, and reviews results.

- **Student:** Attempt quizzes, submits answers, and views results.

The system workflow is represented using UML Use Case Diagrams to illustrate interactions between different system users.

Use Case Diagram:



Use Case Explanation:

The administrator first sets up the system by managing users and permissions. Teachers then create or auto-generate quizzes and assign them to students. During quiz attempts, students interact with the system under active monitoring to prevent cheating. The system evaluates submissions using AI and automated grading logic. Finally, results are stored and made available to students and teachers through dashboards.

This structured workflow ensures secure, transparent, and efficient assessment management.

Chapter 4: Project Planning and Management

4.1) Development Methodology

The proposed project follows the Agile Software Development Methodology, which emphasizes iterative development, continuous feedback, and incremental improvement. Agile is particularly suitable for this project due to the involvement of AI-based components, evolving requirements, and the need for frequent evaluation and refinement.

Process:

The development process is divided into multiple short iterations (sprints), where each sprint focuses on a specific set of features such as system design, quiz management, AI integration, or security mechanisms. At the end of each iteration, the developed components are reviewed and tested, and feedback is obtained from the project supervisor. This feedback-driven approach allows early identification of issues and ensures that the project remains aligned with academic and functional requirements.

Agile methodology also supports flexibility, enabling changes in system design or implementation strategies without disrupting the overall project timeline. This is particularly important for AI-based systems, where model performance and system behavior may require repeated tuning and optimization. Overall, Agile ensures better risk management, improved quality, and timely project completion.

4.2) Gantt Chart

The project timeline is planned semester-wise to ensure systematic progress and timely completion of FYP-I objectives.

The major phases and their durations are represented below:

Phase	Duration
Idea Approval and Proposal	Semester 7, December 2025
Requirement Analysis	Semester 7, December 2025
System Design	Semester 8, February-March 2026
Implementation	Semester 8, April 2026
Testing and Documentation	Semester 8, May 2026
Final Presentation	Semester 8, June 2026

Phase Description

- **Idea Approval and proposal:**

This phase involves approving the selected idea from supervisor and representing the idea and its objectives. The documented proposal is also submitted.

- **Requirement Analysis:**

This phase focuses on understanding the problem domain, identifying system requirements, reviewing existing solutions, and finalizing functional and non-functional requirements.

- **System Design:**

During this phase, the system architecture, database schema, UML diagrams, and workflow designs are developed. Design decisions related to AI integration and security mechanisms are also finalized.

- **Implementation:**

Core system modules such as user authentication, quiz management, AI-based grading, and anti-cheating mechanisms are developed during this phase.

- **Testing and Documentation:**

System testing, bug fixing, performance evaluation, and preparation of final documentation are carried out to ensure readiness for FYP-II.

- **Final Presentation:**

The final website or application is presented to externals and internals and thesis will be submitted.

4.4) Risk Analysis

Risk analysis is essential to identify potential challenges that may affect project execution and to define mitigation strategies in advance.

The major risks associated with this project are represented below:

Risk	Impact	Mitigation Strategy
AI Grading Bias	High	Model fine tuning and validation
Security Bypass	High	Continuous monitoring and updates
Time Constraints	Medium	Agile Planning and task prioritization

Risk Explanation:

- **AI Grading Bias:**

AI-based evaluation models may produce biased or inaccurate grading due to limited training data or linguistic variations in student responses. To mitigate this risk, the system will employ model fine-tuning, testing on diverse sample data, and continuous evaluation to improve grading accuracy and fairness.

- **Security Bypass:**

Since the system is designed to prevent cheating, there is a risk that users may attempt to bypass security controls. This risk will be mitigated through continuous

monitoring, secure authentication mechanisms, regular testing, and activity logging, ensuring early detection of suspicious behavior.

- **Time Constraints:**

Limited semester duration may impact development and testing activities. This risk will be mitigated by Agile planning, clear milestone definition, and prioritization of core features to ensure timely completion of FYP-I objectives.

Chapter 5: Conclusion and Future Work

5.1) Conclusion

This Final Year Project-I successfully establishes a strong foundation for the development of an AI-based intelligent quiz and assessment system with anti-cheating features. The project clearly identifies the limitations of existing online assessment platforms and highlights the need for a secure, automated, and unbiased evaluation system. A comprehensive literature review was conducted to analyze existing solutions, technologies, and research studies, which helped in identifying key research gaps.

Based on the identified problems and gaps, a detailed system architecture and design were proposed, incorporating Artificial Intelligence, Natural Language Processing, and secure web technologies. The functional and non-functional requirements were clearly defined, and an Agile development methodology was selected to ensure flexibility and continuous improvement. Furthermore, a realistic project plan, timeline, and risk analysis were established to ensure smooth execution in subsequent phases.

Overall, FYP-I achieves its objectives by providing a complete conceptual and technical roadmap for the project. The outcomes of this phase ensure that the project is well-structured, feasible, and ready for full implementation in FYP-II.

5.2) Future Work

In FYP-II, the proposed system will be fully implemented, tested, and deployed based on the design and planning completed in FYP-I. The core system modules, including user authentication, quiz management, AI-based question generation, automated grading, and anti-cheating mechanisms, will be developed and integrated into a functional web application.

Comprehensive testing and validation will be carried out to ensure system reliability, performance, and security. This includes unit testing, integration testing, and user acceptance testing to evaluate system behavior under real-world usage scenarios. AI grading models will be fine-tuned using diverse datasets to improve accuracy and fairness.

Future Enhancements:

Several future enhancements can be incorporated to further improve the system:

- **Mobile Application Development:**

A mobile version of the system can be developed to increase accessibility and usability.

- **Learning Management System (LMS) Integration:**

The system can be integrated with existing LMS platforms to support institutional adoption.

- **Advanced Analytics and Reporting:**

Interactive dashboards and data visualization tools can be added for detailed performance analysis.

- **Gamification Features:**

Badges, leaderboards, and rewards can be introduced to enhance student engagement.

- **Advanced Proctoring Techniques:**

Facial recognition and behavioral analysis can be explored for enhanced security.

These enhancements will transform the proposed system into a comprehensive, scalable, and industry-ready assessment platform, contributing significantly to the future of intelligent online education.

References:

A. Brown, “NLP in Educational Technology,” Springer, 2021.